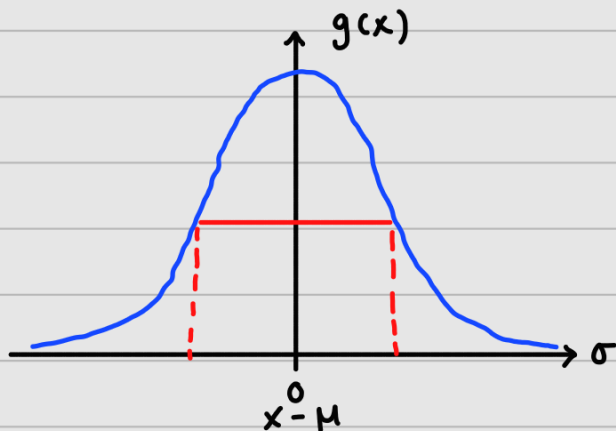


3a

The Gaussian distribution

$$g(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$$

Describes the probability of obtaining the value x in a random observation



Probability that a value lands within an interval

$$dG(x; \mu, \sigma) = g(x; \mu, \sigma) dx$$

The maximum of g occurs at $x = \mu$ (the mean)
So plugging it in

$$g(\mu) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\underbrace{\left(\frac{\mu-\mu}{\sigma}\right)^2}_0\right] = \frac{1}{\sigma\sqrt{2\pi}}$$

$e^0 = 1$

The half maximum occurs at x_F is $\frac{1}{2}$ the max

$$g(x_F) = \frac{1}{2}g(\mu) = \frac{1}{2\sigma\sqrt{2\pi}}$$

Rewritten as

$$\frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x_F-\mu}{\sigma}\right)^2\right] = \frac{1}{2\sigma\sqrt{2\pi}}$$

$$\exp\left[-\frac{1}{2}\left(\frac{x_F-\mu}{\sigma}\right)^2\right] = \frac{1}{2}$$

$\ln($)

$$\frac{1}{2} \left(\frac{x_F - \mu}{\sigma} \right)^2 = \ln\left(\frac{1}{2}\right) = \ln(1) - \ln(2) = -\ln(2)$$

$$\left(\frac{x_F - \mu}{\sigma} \right)^2 = 2 \ln(2)$$

$$\frac{x_F - \mu}{\sigma} = \pm \sqrt{2 \ln 2}$$

$$\begin{aligned} x_F &= \mu \pm \sqrt{2 \ln 2} \sigma \\ &= \mu \pm 1.177 \sigma \end{aligned}$$

This implies there are two points at which the peak value is half the maximum

We can characterize the full width at half maximum as the range between the 2 x_F values

$$\begin{aligned} \Gamma &= \Delta x = x_{F,1} - x_{F,2} \\ &= (\mu + 1.177 \sigma) - (\mu - 1.177 \sigma) \end{aligned}$$

$$\Gamma = 2.354 \sigma$$

#3b

Suppose we have a line on a log-log plot

$$y = mx + b$$

where y values are in $\log y = \log y$

and x values also $x = \log x$

$$\therefore \log y = m \log x + b$$

↓

$$y = x^m 10^b \quad (b = y \text{ intercept})$$

So a straight line in log-log corresponds to

$$y = \beta x^m \text{ where } \beta = 10^b = \text{constant}$$

For the slope between two points

$$\log y_1 = m \log x_1 + b$$

$$\log y_2 = m \log x_2 + b$$

$$\begin{aligned} \log y_2 - \log y_1 &= m \log x_2 + \cancel{b} - (m \log x_1 + \cancel{b}) \\ &= m (\log x_2 - \log x_1) \end{aligned}$$

$$\therefore m = \frac{\log(y_2/y_1)}{\log(x_2/x_1)}$$

And last, the intercept where $x = 0$

$$\log y = b \rightarrow y = 10^b \quad \text{intercept w/ } b = \text{intercept of log-log plot}$$

* Assuming base 10, could be base e like $\ln$ and so on